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Colorless polyimides with excellent optical transparency and self-healing properties based on multi-exchange dynamic network

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ABSTRACT

Colorless polyimides (CPIs) with outstanding mechanical properties are used as essential materials in the production of future flexible display panels, foldable windows, and spacecraft cockpit materials. However, under repetitive stress and deformation, fatigue fractures caused by cracks and radio waves may act as critical determinants of the properties and duration of the material. To solve this problem, the present study developed a CPI capable of self-healing, wherein the transparency and any damage caused by external stress are rapidly and easily restored. Our synthesized CPI films showed ultra-transmittance (> 95%) and excellent self-healing efficiency (> 98%) with a simultaneously induced multi exchange network. Notably, the films exhibited excellent mechanical properties even after a bending fatigue test involving more than 10,000 cycles. This study therefore suggests a new type of CPI with a self-healing function and shows that the trade-off between the durability and functionality can be completely resolved.

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1. Introduction

Polyimides (PIs) are the most representative engineering plastic materials with excellent thermal and chemical resistances as well as outstanding mechanical and electrical properties. They have been used consistently in various industrial fields (aeronautics, dielectrics, electronic devices, etc.) since their first development in the 1960s [1–5]. Aromatic polyimides, however, pose a challenge because of their dark-brown color; they absorb light in the visible spectrum area at approximately 500 nm owing to the charge transfer complex (CTC) within the polyimide molecules and between chains, caused by the conjugated system of π electrons found in the main chain [6]. Therefore, to solve this problem, studies have continuously reported findings on colorless polyimides (CPI), where a monomer is introduced to lower the CTC effect, thereby resolving the color problem and, at the same time, enhancing flexibility [7–12].

The CPI film is referred to as the "*dream material*" in the field of electronic materials, particularly in the display market. The conventional display glass substrate is heavy and easily breakable and does not allow curving or folding, which led to the development of CPIs as a substrate for flexible displays. Because it has a hard surface that resists cracks even after being folded and unfolded hundreds of thousands of times, the film is applied as a display substrate with a high degree of freedom for shapes that allow curving, folding, and bending, with a substantial potential as industrial applications such as electronic devices [4,13,14]. Additionally, the excellent chemical resistance and hardness of the CPI film have led to its use as a high-function, hard-coating material [15,16]. Nonetheless, the weak resistance to scratching implies that the surface has to be treated using a hard coating with a thickness of several micrometers, while the fatigue fracture caused by the cracks and radio waves under conditions of repetitive stress and deformation is a critical determinant of the properties and duration of the material. As a technique to effectively resolve such problems in the field of materials, recent studies have reported the use of active or self-healing polymers based on reversible covalent bonds.

Self-healing polymers employ the concept of a material capable of autonomously and actively detecting damage or flaws caused by various changes in the surrounding environment, and restoring the damage using an external source of stimulus (light, UV, heat, etc.). The method of healing is broadly categorized into the extrinsic (microcapsule [17] and micro-vascular[43]) and intrinsic methods (Diels-Alder, disulfide bond, hydrogen bond, ionic interaction, etc. [18–24]). In particular, self-healing polymers based on reversible covalent bonds enable rapid and proactive healing regardless of the range of the damaged area or repetitive flaws. However,

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Table 1
Composition of the CPI films.

Samples I.D.	Dianhydride	Diamine	Diamine molar ratio(4AD:TTDA)	Molecular weight (10^4 g/mol)	PDI
6FDA0	6FDA (10 mmol)	TTDA (10 mmol)	-	7.3	1.56
6FDA14		4AD (2 mmol) + TTDA (8 mmol)	1:4	7.9	1.65
6FDA32		4AD (6 mmol) + TTDA (4 mmol)	3:2	8.3	1.5
6FDA1		4AD (10 mmol)	-	9.9	1.21

despite being an innovative material technique, self-healing polymers have one major disadvantage: to realize rapid self-healing properties at ambient temperatures (or low temperatures), the material should comprise of a soft material with a low glass transition temperature (Tg) as a prerequisite, which implies that the inherent low mechanical properties of the material should be resolved. Hence, for a material, strong mechanical properties and active self-healing properties are in a tradeoff relationship. Recently, several studies have reported on concurrent complementation where self-healing properties are enhanced while the drawbacks are addressed [25,26].

In general, wholly aromatic polyimides are glassy polymers that exhibit a Tg as high as 200 °C or higher. For this reason, it is extremely difficult for polyimides to exhibit self-healing properties at low temperatures, unlike elastomers. Therefore, to allow glassy polymers, such as polyimides, to exhibit self-healing properties, the Tg must be controlled at low temperatures. Susa et al. used aliphatic diamines of a long chain shape to verify the self-healing potential of polyimides [27,28]. Synthesized PI was healed at room temperature, while maintaining elastomeric-like mechanical properties. However, diamines with aliphatic chains show higher basicity than those of aromatic diamines, which leads to the formation of ionic bonds with dianhydrides at the onset of synthesis, causing them to exist as a salt that ultimately reduces the molecular weight [29–31]. Consequently, they were unable to solve the problem of the reduced mechanical properties, which was caused by the lowering of the Tg as compared to that of the conventional PI.

To overcome the drawbacks discussed above, we propose a novel CPI film that is free from damages caused by repetitive deformations while exhibiting high transparency as well as outstanding thermal and mechanical properties. For this, the Tg of the polyimide was controlled using 1, 12-diaminodecane (DD) and 4, 7, 10-Trioxa-1, 13-tridecanediamine (TTDA) monomers. The problem of the reduced molecular weight caused by the high basicity of aliphatic amines was solved using the bulky trifluoromethyl functional group (-CF_3) of 6FDA. Notably, by adjusting the ratio between 4AD and TTDA and the components of the diamines, the Tg was controlled to allow self-healing properties at relatively low temperatures. Overall, in our system, reversible covalent and non-covalent bonds were simultaneously used in the chain, which resulted in a rapid and perfect restoration as well as sustainable self-healing properties. Ultimately, a CPI film was developed with a high level of transparency and autonomous self-healing properties. The synthesis process is the same for all polyimide samples, and their different processes and designs are presented in Table 1 and Fig. 1(a). The synthesized CPI samples in this study includes four types four types of designs: 6FDA0 for the synthesis based solely on diamines acting as a soft segment (TTDA); 6FDA1 for the synthesis based solely on diamines acting as a hard segment (4AD); 6FDA14 and 6FDA32 for the synthesis based on 4AD and TTDA with a molar ratio of 1:4 and 3:2, respectively.

2. Experimental section

2.1. Materials and synthesis of self-healing colorless polyimide films

Four different polyimides were synthesized using a two-step polymerization process, as described below. The monomers used were 4,4′-(hexafluoroisopropylidene)diphthalic anhydride (6FDA) (99%, Changzhou Sunlight Pharmaceutical Co., Ltd.), 4,4′-dithioaniline (4AD) (98%, Tokyo Chemical Industry), and 4,7,10-Trioxa-1,13-tridecanediamine (TTDA) (97%, Sigma-Aldrich). 6FDA was used as the dianhydride and four kinds of polyimide samples were synthesized by changing the diamine type. Four polymers with different 6FDA/Diamine molar ratios, with respect to the theoretical stoichiometric ratio, were synthesized: 6FDA0 (at the theoretical stoichiometric ratio TTDA(2.20 g, 10 mmol)/6FDA(4.44 g, 10 mmol), 6FDA14 (two types of diamines were used: 0.50 g, 2 mmol of 4AD and 1.76 g, 8 mmol of TTDA), 6FDA32 (two types of diamine were used: 1.49 g, 6 mmol of 4AD and 0.88 g, 4 mmol of TTDA), and 6FDA1 (at the theoretical stoichiometric ratio 4AD(2.48 g, 10 mmol/6FDA (4.44 g, 10 mmol). The synthesis was conducted in an N,N-dimethylacetamide (DMAc, Daejung Chemical) polar aprotic solvent with a total solids (monomers) content of 25 wt%. Diamine (10 mmol) was added to each sample in the DMAc. Dianhydride (10 mmol) was added to the solution. The solution was stirred at 0 °C for 1 h and then overnight (24 h) at room temperature in a nitrogen atmosphere. The polyimide was obtained through the chemical cyclodehydration of polyamic acid (PAA), in which the reaction was terminated. Acetic anhydride (2.55 g, 25 mmol) and pyridine (1.98 g, 25 mmol) were added to the PAA solution and reacted at 50 °C for 10 h. The reaction-terminated solution was precipitated in water that had been stirred at 300 rpm and then washed with methanol to obtain a fibrous polyimide.

2.2. Preparation of self-healing colorless polyimide films

Make a CPI solution of fibrous polyimide in DMAc solvent at a ratio of 30 wt% (solid content). When the fibrous polyimide is completely dissolved in the solvent, the air-bubble is primarily removed by maintaining a vacuum at room temperature. After that, the solution is poured on the glass substrate and spin coating is performed. (Film casting condition : at 1000 rpm for 10 s) Finally, in a vacuum oven at 70 °C for 1 h and at 170 °C for 2 h to remove residual solvent and obtain a CPI film. (average thickness : 30–40 μm)

2.3. Characterization methods

The molecular weight of the synthesized polyimides was determined though gel permeation chromatography (GPC, Thermo Scientific Ultimate 3000) using THF at a concentration of 1 mg/mL. The chemical structures were confirmed through Fourier-transform infrared spectroscopy (FT-IR, Nicolet iS10, Thermo Scientific) at

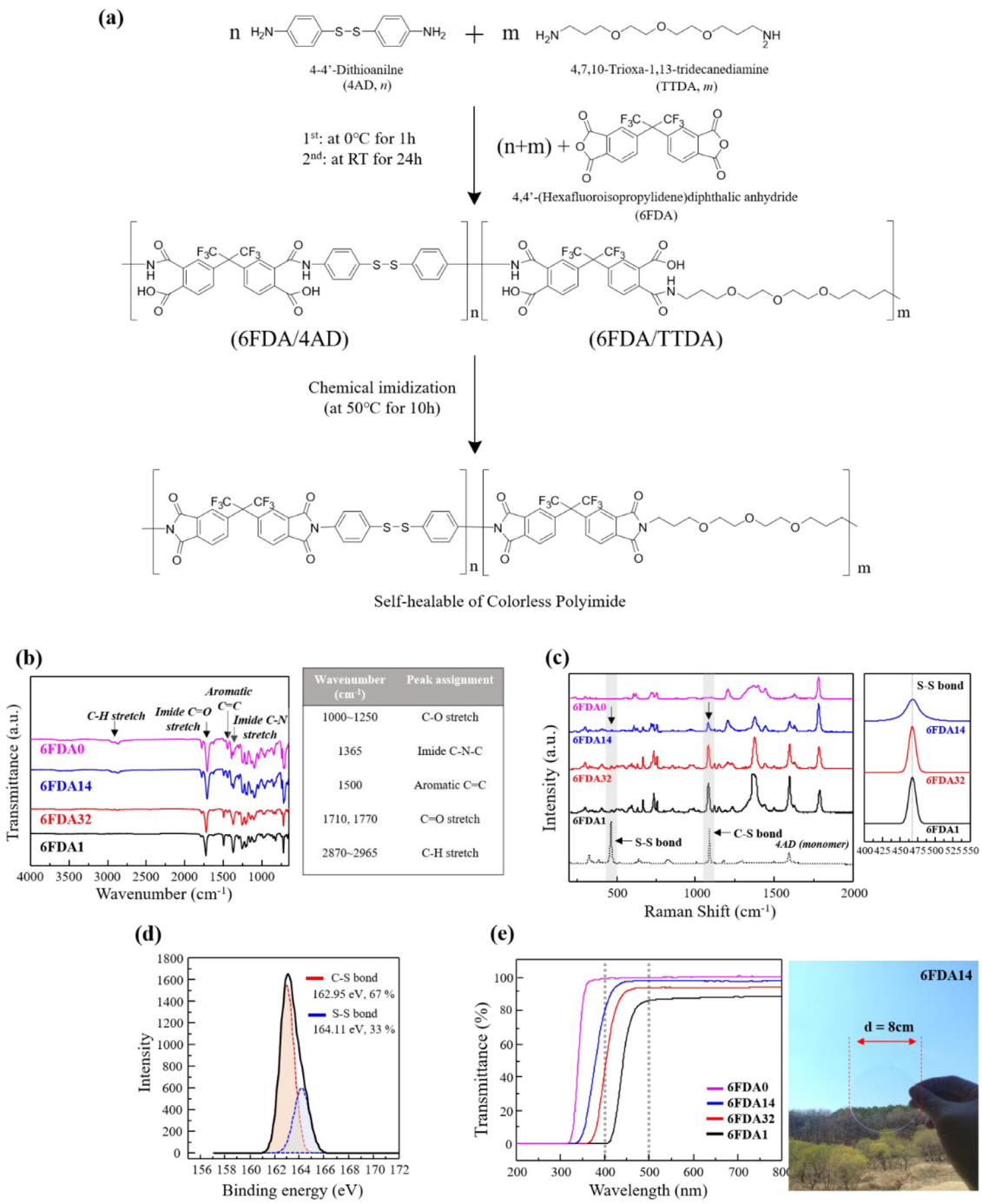

Fig. 1. Structural properties of transparent and flexible self-healing colorless polyimide. (a) Schematic of self-healable polyimide synthesis step, (b) FT-IR spectra, (c) Raman spectroscopy, (d) S2p region in XPS, and (e) UV-vis for the CPI films. (For interpretation of the references to color in this figure, the reader is referred to the web version of this article.)

a resolution of 4 cm^{-1} with 32 scans performed for each measurement.^{1}H-NMR spectra were performed at 25°C on a Agilent NMR 600 MHz. Raman spectra were measured using a Raman spectrometer (Renishaw inVia-reflex Raman microscope) with a 785 nm laser. The UV–vis spectra (Jasco V-670 UV–vis/NIR spectrophotometer) were obtained in transmittance and absorbance mode at 1 nm intervals from 200 to 800 nm. The thermal properties were evaluated through thermogravimetric analysis (TGA, Q50, TA Instruments, USA) and differential scanning calorimetry (DSC, Q20, TA Instruments, USA). The fluorescence spectra of polyimides were obtained using a Olis,Inc.(RMS-1000) spectrofluorometer equipped with 75watt xenon arc lamp and 150 watt xenon arc lamp. Thermogravimetric analysis was used to analyze the thermal degradation of the polyimide films; a 10 mg sample was heated from 35 to 800 °C at a rate of 10 °C /min under a N_2 atmosphere. DSC measurements were carried out at 10 °C/min following this procedure: (1) heating from 35 to 300 °C; (2) cooling down to 0 °C; and (3) repeat steps 1 to 2. The Tg was determined from the second heating cycle. The mechanical properties were determined at room temperature according to the ASTM D638 test method us-

ing a universal tensile machine (UTM, Instron model 5567A) at a load cell of 100 N, gage length of 10 mm, and crosshead speed of 5 mm/min, using dog-bone type dumbbell specimens. The dimensions of the specimens were 30 (length) × 5 (width) × 10 (narrow portion length) × 1.5 (narrow portion width) × 0.03 (thickness) mm. At least 10 samples were tested, and the average of each set of measurements was used. The breaking stress and Young' s modulus of the self-healed samples were obtained from the stress–strain curves. Each result was obtained by repeating the test at least 10 times under the same measurement conditions. To evaluate the dynamic behavior of polyimides, a dynamic mechanical analyzer (DMA, TA Instruments DMA Q800) was used under the condition of a heating rate of 3°C/min at a strain 0.1% (frequncy of 1 Hz), with a 10 μm amplitude using rectanglular samples dimensions of 6 (length) x 7 (width) x 0.05 (thickness)mm. To observe the surface properties of the PI films, we used nano-indentaion (iNano, NANOMECHANICS, Inc.) to evaluate film roughness and X-ray photoelectron spectroscopy (XPS, K-Alpha, Thermo Fisher Scientific, USA) to evaluate the inner structural and surface properties of the fabricated polyimide films in the pristine, cut, and healed states. The healing involved thermal treatment at 125 °C for 20 min and field-emission scanning electron microscopy (FE-SEM, Nova NanoSEM 450, FEI, USA).

For the self-healing tests, the CPI films were damaged by a scratching test machine. After the cutter (DN-52, DORCO) was mounted on the scratching test machine, a constant load was applied to the polyimide film to inflict constant damage. Subsequently, the films were heated on a hot plate for 20 min under different healing conditions, with final temperatures of Tg+10 °C, Tg+20 °C, and Tg+30 °C. Sandpaper (180-grit, commercial sandpaper,) a diamond cutter, and a scalpel blade (#10, FEATHER) cutter were mounted on the scratching test machine, and the polyimide film was damaged under a constant load, after which the healing test was performed.

After scratch and bending traces, the healed CPI films were obtained using a heat gun (Boush GHG 630) at 120 °C for 10 min (the point where the healing of 6FDA14 film begins), as shown in Fig. S5. Following fatigue tests, the CPI films were hard damaged (1R) by a folding tester (Zest Co. Ltd. South Korea) for 10,000 cycles (1 cycle means of 1 Hz/s) as shown in Fig. 5, Fig. S11, and Movies 2 and 3. The dimensions of the specimens were 50 (length) × 20 (width) mm, and 30 (thickness) μm at room temperature and humidity (RH% 20).

3. Results and discussion

Fig. 1(b)–(e) present the structural characteristics and transparency of the synthesized colorless polyimides, and their disulfide bonds as self-healing functional groups within the chain. Fig. 1(b) shows the FT-IR spectra of the CPI prepared by chemical imidization. The characteristic absorption peaks near 1780 cm^{-1}, 1710 cm^{-1}, and 1365 cm^{-1} originate from the in-plane imide group (C=O), out-of-plane imide group (C=O), and axial-imide II (C–N–C) groups, respectively. Moreover, for the imide samples with a mixture of aliphatic diamines, the detection of a peak at 2900 cm^{-1} indicated imide synthesis. It was confirmed through ^{1H}-NMR (Fig. S1) that the imide structure was well formed according to the ratio of the soft segment (TTDA) and hard segment (4AD).

To examine the reversible covalent bond of the disulfide functional group in the synthesized CPI film, measurements were conducted using Raman spectroscopy and X-ray photoelectron spectroscopy (XPS); the respective results are presented in Fig. 1(c)-(d). The results of the Raman spectra measurements for 4AD containing the disulfide bond are shown in Fig. 1(c), wherein the peaks of the disulfide and C–S bonds are visible at 473 cm^{-1} and 1085 cm^{-1}, respectively. In addition, for the CPI (6FDA14, 6FDA32, and 6FDA1) films mixed with 4AD, distinct peaks of the disulfide group were detected at 473 cm^{-1}, where the peak intensity varied according to the change in the composition ratio of 4AD in the film. In other words, 6FDA14, which had the smallest proportion of 4AD, exhibited a relatively low content in comparison with those of 6FDA32 and 6FDA1 with the largest proportion, thereby showing a decreasing trend in the peak intensity. The results are in line with the trend shown by the intensity of the C–S bond peak, and the XPS results also revealed the S–S bond and C–S bond peaks, as shown in Fig. 1(d).

As shown by schematic S1, upon synthesizing the CPIs, the high basicity of aliphatic diamines is known to undergo ionization when dissolved in a solvent. The resulting formation of salt bonds with anhydrides leads to cross-linking between the salt bonds that interfere with the molecular weight increase, which results in the characteristic derivation of low molecular weight. The colorless polyimide film synthesized in this study also had an aliphatic diamine within its structure, which could have contributed to the formation of a film with low molecular weight. However, one of the components, the 6FDA monomer, had a trifluoromethyl functional group (-CF_3), which is thermally stable and a strong electron acceptor, causing an increase in the intermolecular free volume and interfering with the cross-linking of the salt bonds. This resulted in a film with high molecular weight without undergoing special treatments[31]. The GPC results of the synthesized sample in Table 1 support this finding.

The CTC in wholly aromatic PIs absorbs light in the visible ray area at approximately 500 nm, resulting in a yellowish color. Therefore, high transmittance in the visible ray area should be obtained to fabricate CPIs. The thickness of the film used in the measurements was 30 μm, and all $\lambda_\circ$ values were $\leq$ 400 nm, indicating that the transmittance of light began in an area before the visible ray (Fig. 1(e)). Fig. 1(e) shows that, as the disulfide content increased, the transmittance at a specific value of 500 nm steadily decreased from 98 to 85%, although excellent transmittance $\geq$ 80% was exhibited. This is because the characteristic yellowish property of sulfur and the lack of an aliphatic chain with a weak electron-accepting property led to low transmittance. Meanwhile, in the case of 6FDA0 at 500 nm, transmittance $\geq$ 95% was observed as the content of the TTDA monomer increased. This change in properties is due to decreased CT interactions and electronic conjugation originating from the PI chain structures as the content of aliphatic diamine increased[7,31]. Ultimately, the 6FDA14 film exhibited transparency as high as $\geq$ 95% at 500 nm. The thermal gravimetric analysis (TGA) and differential scanning calorimetry (DSC) of the samples are described in the Fig. S2.

The morphologies of the 6FDA0 and 6FDA14 were compared on the bases of FE-SEM measurements, as shown in Fig. S3. Fig. S3(a) shows a typical surface morphology of each polyimide specimen at low and high magnifications. As can be seen from the Fig. S3, it was confirmed that there was no microstructure change due to the presence of disulfide bonds.

Based on these results presented in Fig. S4, 6FDA32 was incompletely healed because of its high Tg (minimal temperature at 155 °C); however, 6FDA0 and 6FDA14 with low Tg (within 75 and 95°C) values were completely healed. Conversely, we expected that the 6FDA0 would have incomplete healing due to the absence of disulfide bonds, but, contrary to expectations, showed a somewhat higher healing efficiency; consequently, the healing efficiency was compared with 6FDA14 at various temperature ranges for a more detailed comparison. (Fig. 2(a,b)). To investigate the quantitative healing property of the CPI films, temperature change and time as the two main factors of the healing property were set as Tg + 10 °C,

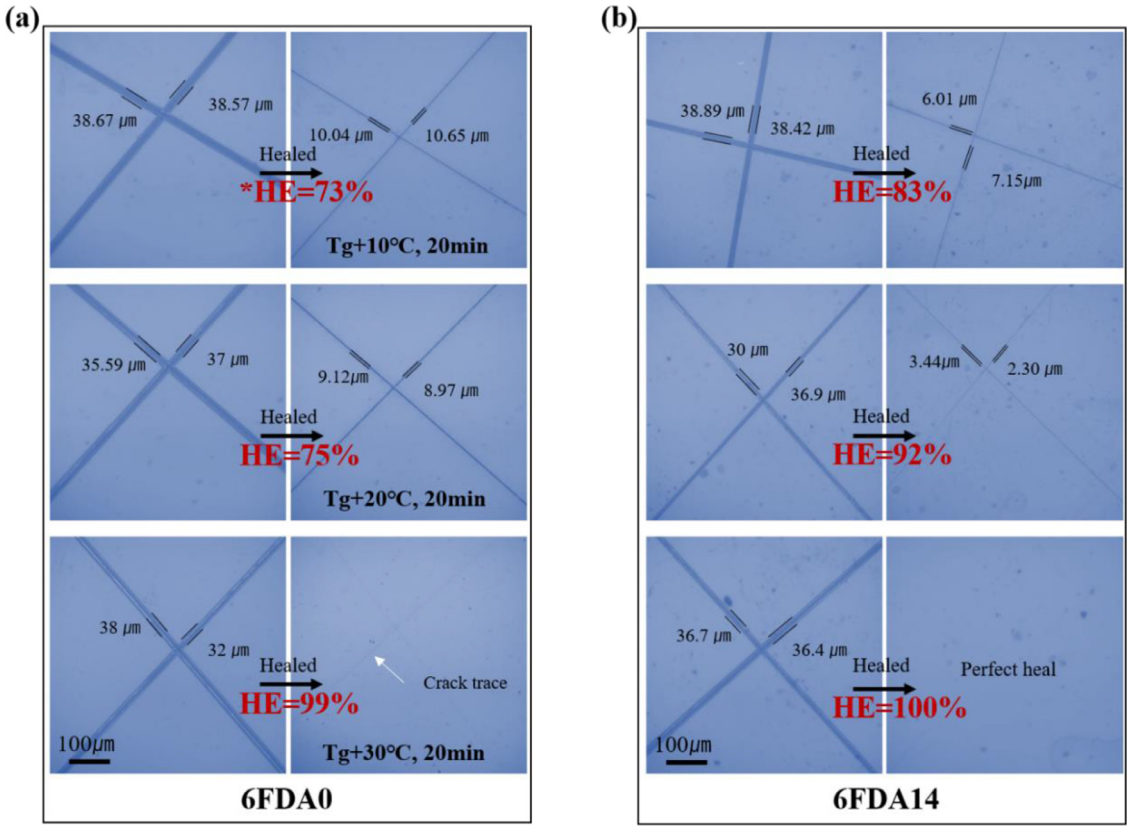

Fig. 2. Self-healing properties of 6FDA0 and 6FDA14 films through optical microscope. (a) Optical microscopy images of before/after healed (a) 6FDA0, and (b) 6FDA14 at Tg+ 10 °C, Tg+ 20 °C, and Tg+ 30 °C for 6FDA14 films, respectively. (*HE=Healing efficiency, Supplementary note 4).

Tg + 20 °C, and Tg + 30 °C and 20 min, respectively, for the evaluation. A scratching test machine developed for this study was used to generate scratches on the film surface. In this machine, a cutter blade (Doroco, Co. Ltd.) of 0.2 mm thickness was used to produce 2 N load, 2 N/min scratch speed, and a maximum scratch length of approximately 600 μm. By applying a constant load on the surface of the CPI film, a cross-shaped scratch was produced, and the temperature was increased from 10 to 30 °C based on the Tg for the healing temperature range, which was maintained at each phase for 20 min. The self-healing properties of the films were subsequently evaluated.

Fig. 2(a) shows 6FDA0 without the disulfide bond as the self-healing functional group, which exhibited an optical self-healing efficiency within 73 and 83% against the initial damage at Tg + 10 °C and 20 min, respectively. In the case of Tg + 30 °C, the healing property was such that the crack trace partially remained but was difficult to verify through visual examination (Fig. 2(a)). In contrast, the 6FDA14 film showed an outstanding healing efficiency of $\geq$ 93% compared with 6FDA0 at 10 °C lower than Tg + 30 °C (i.e., Tg + 20 °C), as shown in Fig. 2(b). For Tg + 30 °C, perfect restoration was observed, which is thought to be due to the disulfide bond (a reversible covalent bond) in the chain. The results indicated that the incorporation of the disulfide bond further enhanced the self-healing function of the synthesized CPI film. However, increasing the composition ratio of the disulfide bond beyond the necessary level may increase the Tg due to an increase in the content of the aromatic ring groups in the chain. As a result, an increase in the self-healing temperature could limit the application scope of the PI film. The results indicate the importance of an appropriate composition ratio for the PI samples.

We observed depth change of damaged 6FDA0 and 6FDA14 specimens by FE-SEM and Micro-CT as show in Figs. S5 and S6. Fig. S5 shows, we analyzed the internal structure of the samples using 3D C-Scan (Micro-CT), a nondestructive analysis technique, to evaluate the scratch of the depth area for the CPI sample, and healed films did not observe any scratch. These results can be easily observed by SEM (Fig. S6). As can be seen from the micro-CT image, it can be seen that the damaged area is uniformly restored according to the healing conditions. Fig. S6 shows, the cross-section SEM images according to the different healing conditions after making a cut as much as half the thickness of the specimen. In case of Tg+ 10°C condition, the 6FDA0 film has a lower damage recovery rate than the 6FDA14. In the case of Tg+ 10°C, the internal damage of both specimens is restored, but damage marks remain on the surface of 6FDA0.

To examine the mechanical properties of the self-healing CPI film and the hardness of the samples, a tensile test, nanoindentation and pencil hardness evaluation were conducted as shown in the supporting Information (Figs. S7–S9 and Table S1). In Fig. 3(a,b), the mechanical properties of the 6FDA0 and 6FDA14 films were compared depending on the presence of the disulfide bond, before and after self-healing, to assess the correlation between the structure and properties. As shown in Fig. 3(c), both samples showed 50–60% healing efficiency in the Tg + 10 °C phase. However, as the healing temperature was increased by 10 °C, the mechanical healing efficiency of 6FDA0 was 70%, whereas that of 6FDA14 was $\geq$ 90%. This can be understood as an identical trend compared with the results of the scratching test for the film surface in Fig. 2. Dynamic mechanical analysis (DMA) was related to the thermal transition of the microdomains in PI. The T_g and vis-

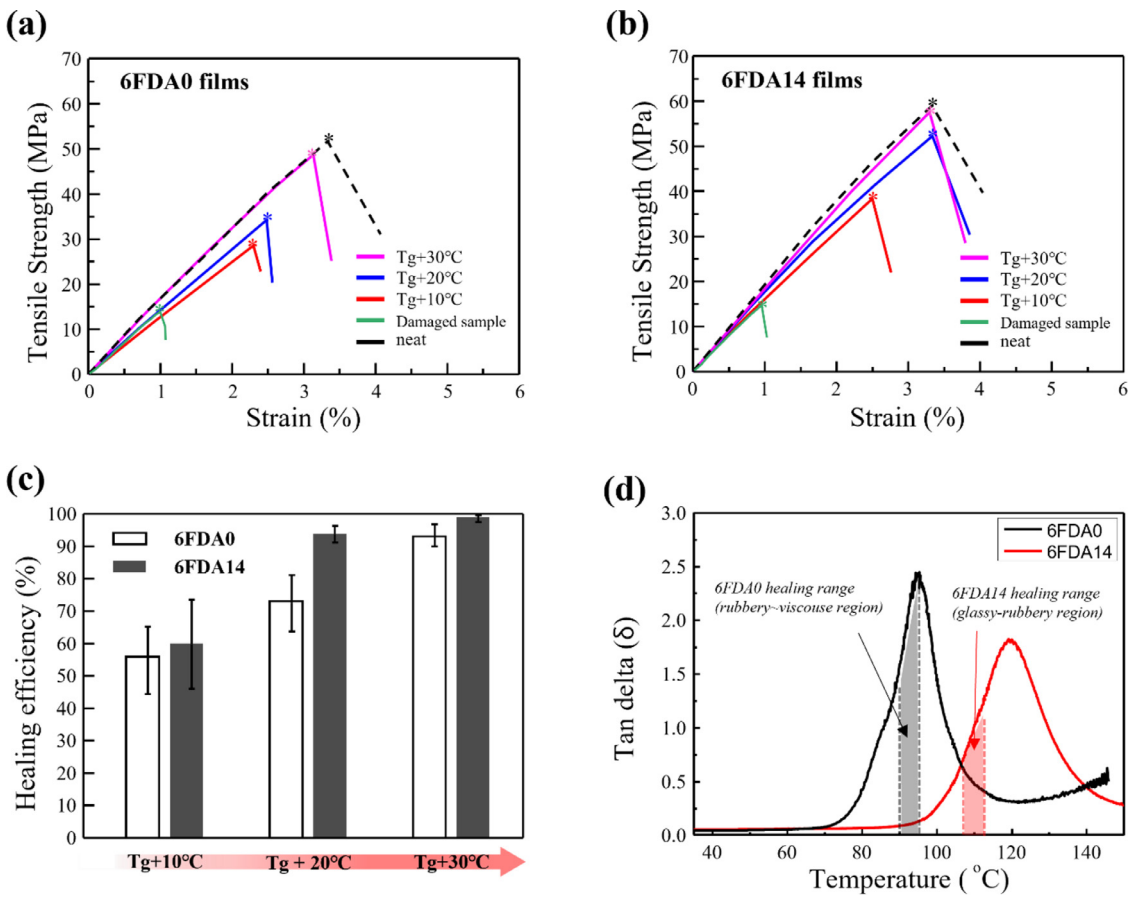

Fig. 3. Stress-stain curves for the before/after healed (a) 6FDA0, (b) 6FDA14, (c) Comparing the self-healing efficiency of 6FDA14 and 6FDA0 (d) DMA results of 6FDA0 and 6FDA14. (*Break point of each specimen).

couse transition temperature of the 6FDA0 and 6FDA14 films were measured using tan δ in DMA as a function of the temperature, as shown in Figs. 3(d), S2(b) and S10. In the case of 6FDA0, T_g indicated by the tan δ peak was observed at approximately 95 °C, 30 °C lower than that of 6FDA14. This result is the same as that of the previously confirmed DSC result. Moreover, considering the healing temperature range based on the scratch and tensile test recovery results (Fig. 3(a)–(d)) of each specimen, it can be inferred that 6FDA0 heals in the rubbery∼viscouse region, while 6FDA14 heals in the glassy–rubbery region. In other words, in the case of 6FDA0, the self-healing efficiency increases as the maximum Tg value of tan δ is approached, whereas in the case of 6FDA14, sufficient recovery characteristics are observed at a temperature lower than Tg.

Self-healing polymers self-heal by chain movement when energy is supplied to the area where the polymer chain is damaged by external physical stimuli (Temp, UV, etc.). However, there are two conditions that need to be fulfilled: first, there must be elements capable of restoring intra-chain bonds or attraction, and second, for these elements to be able to act, the damaged area must be attached or have sufficient polymer chain mobility. The fluidity (mobility) is better at higher temperatures than the fluidity at lower temperatures, and because of this, one can expect a high speed and high self-healing efficiency [25,32]. However, in this study, as disulfide bonds are added, 6FDA14 exhibits a higher Tg and storage and loss moduli than those exhibited by 6FDA0, resulting in low polymer chain fluidity. Nevertheless, the reason for the high self-healing efficiency is that even in the low chain mobility range where self-healing begins, the reversible covalent bonding effect is exerted by the disulfide bond at the same time when the chains start to move, thereby enhancing the self-healing efficiency.

From these results, we can propose several factors that affect the self-healing properties of the synthesized transparent polyimide. The schematic of Fig. 4(a) shows the chemical structure of the synthesized transparent CPI and depicts the self-healing effect that exists between the imide chains. To understand the interaction between the imide chains, photoluminescence (PL) and ultraviolet–visible (UV–vis) spectroscopies were performed, and the self-healing mechanism was confirmed through XPS. (Fig. 4(b)–(d))

The first is the disulfide bond effect, which is a reversible covalent bond (disulfide bond, S–S). Disulfide bonds can be broken by external stimuli, and free radicals exist in broken sulfur to form new disulfide bonds [32–34]. To confirm the disulfide bond healing mechanism, a micro-diameter scratch was generated on the surface of the 6FDA14 film, and the structural property changes before and after healing were confirmed by XPS (Fig. 4(b)). In Fig. 4(b), only C–S and S–S peaks appeared in the spectrum of the film before scratching. On the other hand, in the spectrum of the damaged region, C–S and S–S peaks appeared in two regions owing to the difference in binding energy between the newly formed disulfide bond and the existing disulfide bond. However, as shown healed region, only C–S and S–S peaks appeared because of the formation of newly arranged disulfide bonds in the region where healing occurred completely [35–37]. The second is about the dipole–dipole effect, and it can be expected that strong intra-chain bonds may

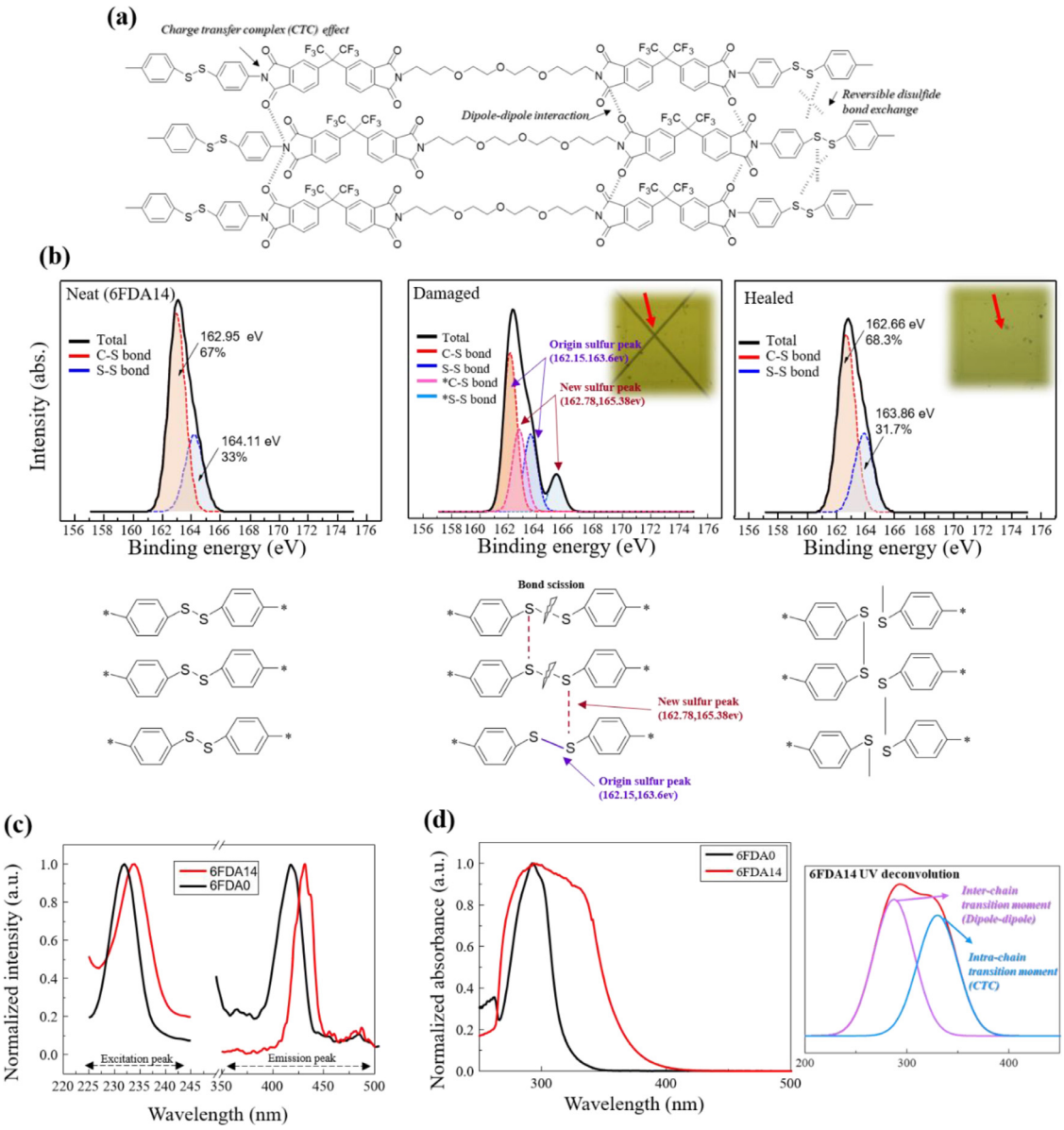

Fig. 4. (a) Schematic representation of the healing based on a multi effect. (b) S2p region in X-ray photoelectron spectroscopy (XPS) for 6FDA14 healing mechanism. (c) PL spectra for emission and excitation peak for 6FDA14, 6FDA0. (d) UV-vis absorption of 6FDA14. 6FDA0 (outside graph 6FDA14 deconvolution).

exist, as shown in Fig. 4(a). To prove this, we confirmed the dipole–dipole interaction via PL and UV–vis absorption peaks (Fig. 4(c,d)) [28,38,39].

In Fig. 4(c), it can be seen that emission and excitation corresponding to the dipole–dipole interaction appear at 232 and 420 nm, respectively. In particular, in the case of 6FDA14 containing disulfide bonds, compared with 6FDA0, it can be understood that the red shift is approximately 10 nm because of conjugate effect difference in the chain. In addition, the UV-vis absorption peak at $\lambda abs < 300$ nm can be confirmed, and both samples show strong absorption peaks caused by the dipole–dipole interaction in the same region at approximately 280 nm. However, in the case of 6FDA14, the absorption peak is confirmed to be in the range of 350 nm. By separating the UV-vis range from 280 to 350 nm for the 6FDA14 sample, a strong peak at 350 nm caused by the CTC effect can be confirmed in addition to the dipole–dipole interaction (Fig. 4(d)). In general, CTCs are capable of intermolecular or intramolecular interactions through electron transfer effects in fully aromatic imides, but the imide synthesized in this study has a semi-aromatic structure; therefore, it is difficult to achieve much better results. However, in the case of 6FDA14, a partial CTC interaction in the chain can be expected because of the composition of 4AD in the chain [38]. In other words, the key outcome of the polymer design in this study is that we could develop a new type of dynamic bond that can cause self-healing. Finally, the synthesis process of the control sample (6FDA-DD) is depicted in Fig. S11 to understand how the introduction of the ether functional group affects the fluidity and self-healing effect of the polymer chain. Fig. S12 is used to examine the effect of chain mobility on self-healing. As shown in Fig. S12(a), after cross-shaped scratches were made on the 6FDA-DD film, healing proceeded at Tg + 30 °C, but the crack trace was clearly visible. As shown in Fig. S12(b), it was found through the mechanical test that the healing efficiency reached approximately 60%. From Fig. S12(c,d), it can be confirmed that the interaction between the polymer chains occurs through the dipole–dipole moment, but the 6FDA-DD sample lacking chain mobility is not completely healed. The reason is that the diamines used in 6FDA-DD have a hydrocarbon structure with-

Table 2
Summary of mechanical properties of CPI film according to healing conditions.

Samples I.D.	Healing condition(°C and min)	Tensile Strength(MPa)	Young's modulus(MPa)
6FDA0	Damaged sample	12 ± 1	1300 ± 468
	Tg+ 10°C, 20 min	28.16 ± 10.8	1252.5 ± 493.9
	Tg+ 20°C, 20 min	36.7 ± 5.7	1583.7 ± 102
	Tg+ 30°C, 20 min	46.9 ± 5.5	1595.3 ± 123.46
6FDA14	Damaged sample	15.28 ± 1.5	1700 ± 308
	Tg+ 10°C, 20 min	34.02 ± 11.43	1640.3 ± 157.9
	Tg+ 20°C, 20 min	53.9 ± 6.54	1818.0 ± 146.4
	Tg+ 30°C, 20 min	56.87 ± 14.7	1945.5 ± 308

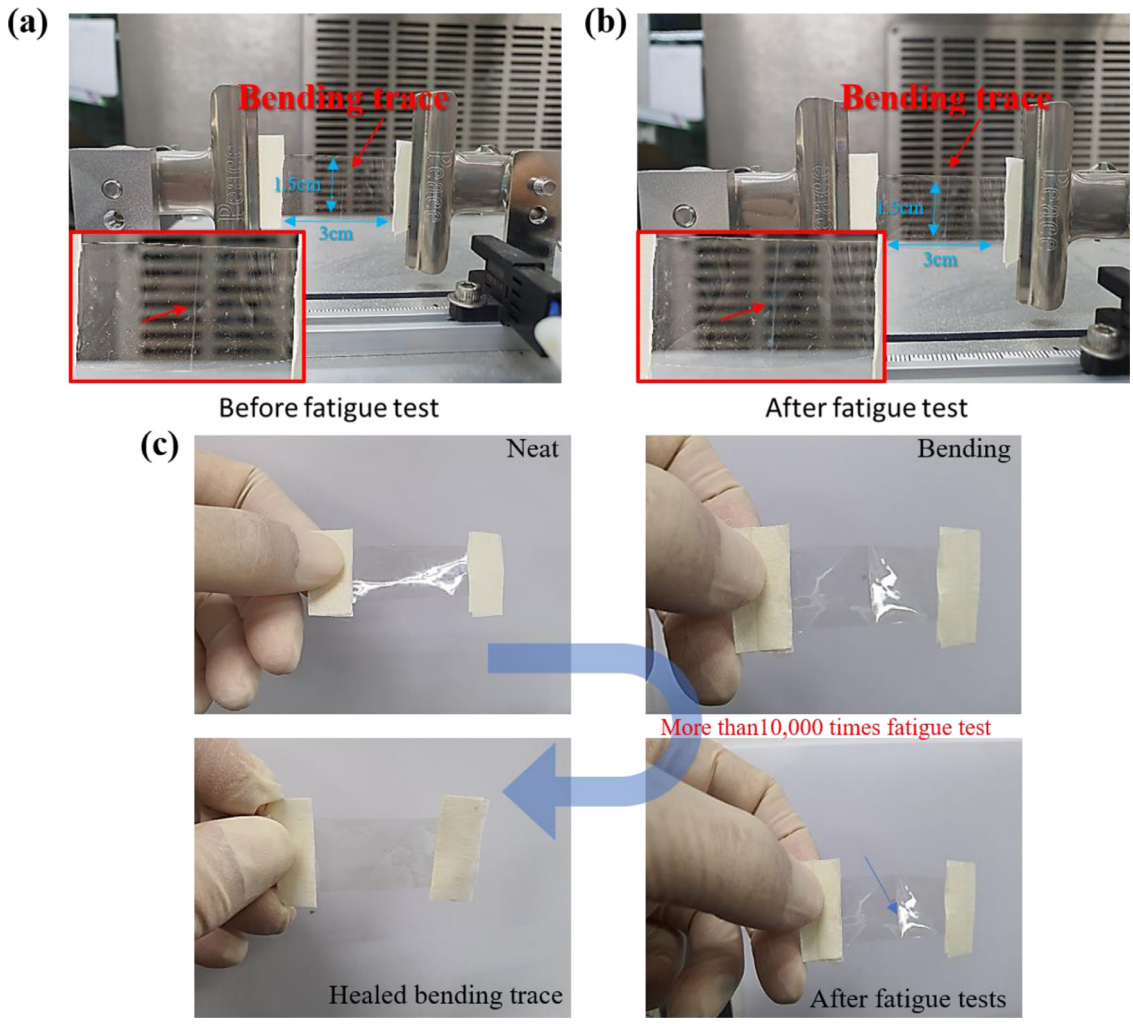

Fig. 5. Photography images for bending test of 6FDA14 with bending trace: (a) Before bending test. (b) After bending test. (Inside photography is an enlarged part of bending trace) (c) before/after healed 6FDA14 film for bending test.

out ether functional groups, which significantly reduces their mobility. In summary, the self-healing mechanism of PI used in this study increased chain mobility by the introduction of the ether (–O–) functional group, and facilitated the disulfide bond formation, dipole–dipole moment, and CTC effects directly involved in self-healing showed high self-healing efficiency (Table 2).

Moreover, after a bending fatigue test involving more than 10,000 fatigue cycles, the material did not easily break or get damaged (Experimental section, Figs. 5, S13 and S14, Movies S2 and S3). In addition, the damaged area could be completely restored using a hair dryer, which is common in homes, when the deformation of the film was caused by fatigue, such as repetitive folding.

4. Conclusions

In this study, CPI films with low Tg values were synthesized using diamines with an aliphatic chain. The Tg of CPI was controlled by adjusting the content of the 4AD monomer containing disulfide bonds and acting as a hard segment. Furthermore, the presence of the ether functional group in the TTDA monomer (building block) acting as a soft segment led to enhanced mobility of the polymeric chain, thereby increasing the healing efficiency. Based on these findings, a CPI film (6FDA14) with outstanding properties such as high transmittance (> 95%), self-healing efficiency (> 98%), and excellent hardness (100 MPa) was synthesized. The market for electronic devices and displays with a flexible function, includ-

ing foldable smart phones and flexible electronic devices, has recently shown rapid expansion. Moreover, the application of the most widely used transparent imides has grown proactive. Because the CPIs developed in this study have self-healing properties, it is anticipated that they will prove useful in various industrial fields.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Young Nam Kim: Writing – original draft. **Juheon Lee:** Investigation. **Young-O Kim:** Investigation. **Jaewoo Kim:** Validation. **Haksoo Han:** Writing – original draft. **Yong Chae Jung:** Writing – original draft.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.apmt.2021.101226.

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